

CoreOps Week 2 Thu Module

Kubernetes Internals & High Availability Architecture

Objective: Master control plane internals, node lifecycle, upgrade patterns, HA design, static pod behaviour, and failure-handling logic.

THURSDAY: Upgrades, version skew, static pods

Critical Concepts

- Kubernetes version skew policy
- Upgrade sequence (API server → CM → Scheduler → Kubelet)
- Why control plane upgrades always come first
- Static pod mechanics
- Static pod failures & drift

Key Outcomes

- You can plan a full enterprise-grade upgrade
- You understand failure blast radiuses
- You can debug static pod corruption

WHAT TO PRACTICE (HANDS-ON)

Lab 1: Modify the API server manifest incorrectly

Add wrong flag → watch API server break.

Lab 2: Change kubelet version (simulate skew failure)

Lab 3: Fix static pod drift

Move a manifest → observe automatic recreation.

WHERE TO STUDY

- Version Skew Docs:
<https://kubernetes.io/docs/setup/release/version-skew-policy/>
 - Upgrade Guide: K8s official docs
 - Kubeadm upgrade flow
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INFRATHRONE MINDSET NOTES

- Production upgrades fail because of version skew.
 - Static pods are the spine of your control plane; if they break, your cluster breaks.
 - Never upgrade blindly; always rehearse upgrades in a clone cluster.
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